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Professional Summary:

- 5+ years of experience in Data Engineering with hands on experience in building data intensive applications and creating pipelines using Python and Shell scripting with good experience in Cloud and Bigdata Technologies.
- Good experience with Data Extraction, Transformation and Loading (ETL). Building a Data warehouse using Star and Snowflake schemas. Well versed with scrum methodologies.
- Proficient in writing Bash, Perl, Shell and Python scripts to automate and provide Control Flow.
- Experience in developing web applications and implementing Model View Control (MVC) architecture using server-side applications Django and Flask.
- Excellent experience with NumPy, Matplotlib, SciPy, PySpark and Pandas python libraries during development lifecycle and experience in developing API's for the application using Python, MongoDB, Express, ReactJS, and NodeJS.
- Experience in Database Design and development with Business Intelligence using SQL Server Integration Services (SSIS), SQL Server Analysis Services (SSAS), OLAP Cubes, Star Schema and Snowflake Schema.
- Strong experience with Conceptual, Logical, and Physical Data Modelling skills, Data Profiling skills, maintaining data quality, experience with JAD sessions for requirements gathering, creating data mapping documents, and writing functional specifications, and queries, Dimensional Data Modelling, FACT & Dimension tables.
- Deep knowledge and strong deployment experience in Hadoop and Big Data ecosystems- HDFS, MapReduce, Spark, Sqoop, Hive, Kafka, zookeeper, and HBase.
- Proficient in AWS services like VPC, Cloud front, EC2, ECS, EKS, Elastic Beanstalk, Lambda, S3, Storage gateway, RDBS, DynamoDB, Redshift, Elastic Cache, DMS, SMS, Data Pipeline, IAM, WAF, Artifacts, API gateway, SNS, SQS, SES, Auto Scaling, Cloud Formation, Cloud Watch, and Cloud Trail
- Experience in optimizing Volumes and EC2 instances and created multiple VPC instances. Deployed applications on AWS using Elastic Beanstalk and Implemented and set up Route53 for AWS Web Instances.
- Expertise in writing PySpark job in AWS Glue to merge data from multiple tables and in Utilizing Crawler to populate AWS Glue Data Catalog with metadata table definitions.
- Experience working with systems processing event based logs, user logs, machines, Azure AD, Azure Data Factory, event hub and event Queue etc.
- Experience in building the Orchestration on Azure Data Factory for scheduling purposes.
- Experience in implementing Security in Web Applications using Azure and deployed Web Applications to Azure.
- Experience working with Azure Logic App Integration tool with different triggers
- Experience in Orchestrated data integration pipelines in ADF using various Activities like Get Metadata, Lookup, For Each, Wait, Execute Pipeline, Set Variable, Filter, until, etc.
- Experience in data processing and transformations and actions in Spark by using Python (PySpark) language.
- Expertise in using Spark Data Frames, Spark-SQL and RDD API of Spark for performing various data transformations and dataset building.
- Experience in developing RESTful web Services to retrieve, transform and aggregate the data from different end points to Hadoop (HBase, Solr).
- Experienced in troubleshooting ETL jobs, Data warehouse, and Data mart data store models.
- Strong experience in data engineering and building ETL pipelines on batch and streaming data using PySpark, Spark SQL
- Experience in creating real time data streaming solutions using Apache Spark/Spark Streaming/Apache Storm, Kafka and Flume.
- Experience working on Spark applications extensively using Scala as the main programming platform. Processing this data using Spark Streaming API with Scala.
- Expert in developing Python-based API - RESTful Web Service to track revenue and perform revenue analysis.
- Hands-on experience in Spark Core, Spark SQL, Spark Streaming and Spark machine learning using Scala and Python programming languages.
- Good experience in building and optimizing big data pipelines, architectures and data sets on building Data Lakes.
- Experience in developing large scale data pipelines using Spark and Hive.
- Strong experience with spark real time streaming data using Kafka and Spring Boot API.
- Experience working with Kafka Zookeeper for managing the cluster with broker/cluster technology.
- Experience in constructing and manipulating large datasets of structured, semi-structured, and unstructured data and supporting systems application architecture using tools like SAS, SQL, Python, R, Minitab, Power BI, and more to extract multi-factor interactions and drive change.
- Experience in handling performance tuning, conducted backups, and ensured the integrity and security of databases managed by Postgres DB in the AWS environment and Aurora-Postgres.
- Experience in delivering zero defect code for three large projects which involved changes to both the front end (web services) and back end (Oracle, Teradata).
- Experience in publishing Power BI Desktop reports created in Report view to the Power BI service.
- Expertise in creating Jenkins Pipeline using Groovy scripts for CI/CD.
- Experience working with different databases Oracle, SQL Server, Teradata, and Cassandra SQL Programming.
- Proficient in writing UNIX shell scripts to process Data Warehouse jobs, file operation purposes, and data analytics purposes.

Technical Skills:

Programming & Scripting	SQL, PySpark, Python, Scala, Bash, Shell, Perl
AWS Services	CloudWatch, Route 53, Elastic Beanstalk (EBS), Auto Scaling, Security Groups, EC2 Container Service (ECS), Dynamo DB, Auto Scaling, Security Groups, Redshift, CloudWatch, CloudFormation, DynamoDB
Azure Services	Azure Data Bricks, Azure Data Factory (ADF), Blob Storage, Azure SQL, Azure Data Lake
Python Frameworks	Django, Flask
Python Libraries	NumPy, Matplotlib, SciPy, PySpark, Pandas
Version Control	Git, Bitbucket
Visualization Tools	Seaborn, QuickSight Power BI, Tableau, Matplotlib.
Big Data Technologies	Scala, Spark, Kafka, Nifi, Airflow, Flume, Snowflake, HDFS, Map Reduce, Pig, Hive, Sqoop, Oozie
Hadoop Frameworks	Cloudera CDHs, Hortonworks HDPs
Data Modelling Schemas	Star Schema, Snowflake Schema
BI Tools	Tableau, Power BI
Database	MySQL, Oracle, MS-SQL Server, Teradata, HBase, Cassandra, Dynamo DB, Mongo DB
Operating Systems	Windows 10, Windows 8, Windows 7, Linux, Unix, Mac OS-X, CentOS
Methodologies	Agile, Waterfall

Professional Experience:**JPMorgan Chase & Co., Tampa, FL.****Jan 2023 - Present****Sr. Data Engineer****Responsibilities:**

- Worked on collection of large sets of Structured and Unstructured data using Python Scripting.
- Created web-based applications using Python on Django framework for data processing.
- Designed and developed data ingestion, aggregation, and integration in the Hadoop environment.
- Implemented the pre-processing procedures along with deployment using the AWS services and creating virtual machines using EC2.
- Created Spark data frames using Spark SQL and prepared data for data analytics by storing it in AWS S3.
- Responsible for loading data from Kafka into HBase using REST API.
- Developed the batch scripts to fetch the data from AWS S3 storage and perform required transformations in Scala using Spark framework.
- Integrated Apache Airflow with AWS to monitor multi-stage ML workflows with the tasks running on Amazon Sage Maker.
- Created, configured the Redshift scalable clusters that supports the data warehouse and created databases on the cluster
- Deployed the application using Docker and AWS Console services.
- Used Spark streaming APIs to perform transformations and actions on the fly for building a common learner data model which gets the data from Kafka in near real time and persists it to the HBase.
- Used Apache Kafka and Spark Streaming to get the data from adobe live stream rest API connections.
- Created Sqoop scripts to import and export customer profile data from RDBMS to S3 buckets.
- Created a deployment document for the developed code and provide support during the code migration phase.
- Worked on data extraction, aggregations and consolidation of Adobe data within AWS Glue using PySpark
- Extracted, Transformed and Loaded data from Sources Systems to Azure Data Storage services using a combination of Azure Data Factory, T-SQL, Spark SQL, and U-SQL Azure Data Lake Analytics.
- Worked on Data Ingestion to one or more Azure Services - (Azure Data Lake, Azure Storage, Azure SQL, and Azure DW) and processing the data in Azure Databricks.
- Coordinated with the application support team and helped them assist understand the business and necessary components for the Integration, Extraction, Transformation and load data.
- Developed PySpark programs and created the data frames and worked on transformations
- Installed and configured Apache Airflow for S3 bucket and Snowflake data warehouse.
- Documented logical, physical, relational, and dimensional data models. Designed the Data Marts in dimensional data modelling using Star and Snowflake schemas.
- Validated the data to check for the proper conversion and identifying and cleaning unwanted data, data profiling for accuracy, completeness, consistency.
- Prepared standard reports, charts, graphs, and tables from a structured data source by querying data repositories using Python and SQL.
- Developed and produced a dashboard, key performance indicators and monitor organization performance.
- Utilized Power BI to create various analytical dashboards that help business users to get quick insight of the data.
- Ingested data from various source systems like Teradata, MySQL, Oracle databases.

Environment: Python, Django, AWS EC2, S3, Sage Maker, Redshift, AWS Glue, Azure Data Lake, Azure Storage, Azure SQL, Azure DW, Databricks, Star Schema, Snowflake Schema, Hadoop, Scala, Sqoop, HBase, Hive, Spark, Spark SQL, Kafka, REST

API, Apache Airflow, PySpark, SQL, Power BI

Symplr, Bangalore, India.

Aug 2021- July 2022

Data Engineer

Responsibilities:

- Worked on server-side applications using Python programming.
- Developed Python based API (RESTful Web Service) using Flask, SQLAlchemy and PostgreSQL.
- Involved in designing and deploying multi-tier applications using all the AWS services like (EC2, Route53, S3, RDS, Dynamo DB, SNS, SQS, IAM) focusing on high-availability, fault tolerance, and auto-scaling in AWS Cloud Formation
- Implemented AWS lambda architectural model for handling end-to-end real time and batch analytic loads.
- Applied various data mining techniques: Linear Regression & Logistic Regression, classification, clustering.
- Published IOT Data to Kafka Stream, Consumed by a spark module to perform predictive analysis using Random Forest Machine learning algorithm.
- Facilitated deployment of multi-clustered environments using AWS EC2 and EMR apart from deploying Dockers for cross-functional deployment.
- Worked on design and deployment of Hadoop cluster and different big data analytic tools, including Pig, Hive, Oozie, Zookeeper, Sqoop, Flume, Impala, Cassandra with Horton work distribution.
- Created Hive external tables on top of data from HDFS and wrote ad-hoc hive queries to analyze the data based on business requirements.
- Utilized Partitioning and Bucketing in Hive to improve hive query processing times.
- Performed incremental data ingestion using Sqoop as an existing application is generating data on a daily basis.
- Migrated and implemented Map Reduce jobs to Spark applications for better performance.
- Designed object model, data model, tables, constraints, necessary stored procedures, functions, triggers, and packages for Oracle Database.
- Worked on Spark applications for data validation, cleansing, transformations, and custom aggregations.
- Developed custom aggregate functions using Spark SQL and performed interactive querying.
- Responsible for installing cluster, commissioning & decommissioning of Data node, Name node high availability, capacity planning, and slots configuration.
- Wrote Python code to schedule jobs dynamically in Apache Airflow.
- Developed Spark applications for the entire batch processing by using Scala.
- Stored the time-series transformed data from the Spark engine built on top of a Hive platform to Amazon S3 and Redshift.
- Encoded and decoded JSON objects using PySpark framework to create and modify the data frames in Apache Spark.
- Visualized the results using Tableau dashboards and the Python Seaborn libraries were used for Data interpretation in deployment.
- Built Elastic Search Cluster in integration with Kibana for publishing real time dashboards for maintenance data.
- Worked on Power BI reports using multiple types of visualizations including line charts, doughnut charts, tables, matrix, KPI, scatter plots, box plots, etc.
- Created a deployment document for the developed code and provide support during the code migration phase.
- Created an initial unit test plan to demonstrate that the software, scripts, and databases developed conform to the design document.
- Provided support during the integration testing and User Acceptance phase of the project.

Environment: Python, Flask, Seaborn, AWS EC2, S3, Elastic Search, Sage Maker, Redshift, SNS, SQS, IAM, AWS Glue, Star Schema, Snowflake Schema, Hadoop, Scala, Sqoop, HBase, Hive, Spark, Spark SQL, Kafka, REST API, Apache Airflow, PySpark, SQL, SQLAlchemy, PostgreSQL, Power BI

ACL Digital, Hyderabad, India

Jan 2019- July 2021

Data Engineer

Responsibilities:

- Responsible for gathering requirements, system analysis, design, development, testing and deployment.
- Created end to end Data pipelines using ADF services to load data from On-premises to Azure SQL server for Data orchestration.
- Implemented Azure Logic apps, azure functions, azure storage, and service bus queries for large enterprise level ERP integration system.
- Built complex ETL jobs that transform data visually with data flows or by using compute services Azure Databricks, and Azure SQL Database
- Worked on data pipeline using Azure Service like Data Factory to load the data from Legacy, SQL server to Azure Data warehouse using Data Factories and Databricks Notebooks.
- Used various activities like data movement activities, transformations, and control activities and copy data, Data flow, Get Metadata, Lookup, Stored procedure, Execute Pipeline.
- Created Pipeline's to extract data from on premises source systems to Azure Cloud Data Lake Storage.
- Worked on Azure Data Factory activities such as Lookups, Stored procedures, if condition, for each, Set Variable, Append Variable, Get Metadata, Filter and wait.
- Implemented Performance tuning techniques in Azure Data Factory and Synapse.
- Extensively used Azure Key Vaults to configure the connections in linked services.

- Configured and implemented the Azure Data Factory Triggers and scheduled the Pipelines. Created and configured a new event hub with the provided event hubs namespace.
- Worked on ARM templates to deploy in production using Azure DevOps. Implemented in creating systems processing event-based logs, user logs, etc.
- Analyzed data where it lives by Mounting Azure Data Lake and Blob to Databricks.
- Maintain and worked with data pipeline that transfers and processes several terabytes of data using Spark, Scala, Python, Apache Kafka, Pig/Hive and Impala.
- Used Oozie workflow engine to manage interdependent Hadoop jobs and to automate several types of Hadoop jobs such as Java MapReduce, Hive and Sqoop as well as system specific jobs.
- Created and configured a new event hub with the provided event hubs namespace.
- Worked on PySpark and Spark SQL transformation in Azure Databricks to perform complex transformations for business rule implementation
- Worked on Data Governance, i.e., Metadata management, Master data Management, Data Quality, and Data Security.
- Participated in the development improvement and maintenance of Snowflake database applications
- Developed Spark applications using PySpark and Spark SQL for data extraction, transformation, and aggregation from multiple file formats for analyzing and transform.
- Created tables, views, sequences, triggers, table spaces, constraints, and generated DDL scripts for physical implementation.
- Worked on Performance Spark Tuning and Query Optimization
- Created a GIT repository and added the project to GitHub.

Environment: Python, Azure, Azure Data Lake, Azure Data Factory, Azure Synapse, Performance Tuning, ADLS, Azure Logic Apps, Azure Event Hub, Databricks, Hadoop, Scala, Oozie, Apache Kafka, Pig, Hive, Impala, MapReduce, Sqoop, Snowflake, Spark, Spark SQL, PySpark, GitHub

Education:

University of South Florida

Master of Science in Business Analytics and Information Systems

Jawaharlal Nehru Technological University

Bachelor of Technology in Electronics and Instrumentation Engineering

Aug 2022 – May 2024

GPA: 4/4

Aug 2017 – Jul 2021

GPA: 3.6/4